

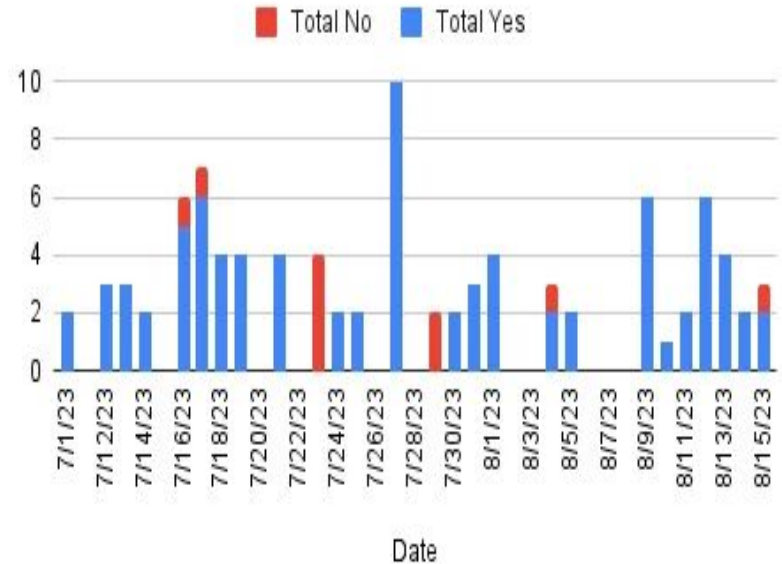
Event attendance forecasting for Diaspora Student communities: the XGBoost approach in resource-limited settings

The problem (Hook)

In our African student community in the Russian Far East, organizing events is fraught with logistical difficulties — long distances, harsh weather conditions, limited transportation. We often order too much food, rent too large rooms, or don't cook enough. The real problem is that we don't know who's actually coming, even if they say yes."

Judging by my posts in the communities, 40-60% of those invited do not come. A simple predictor could save 30% of wasted resources.

RSVPs vs Date - Yes/No Breakdown



Why ML? Why XGBoost?

Requirement

Handles missing data (students forget to answer pre-event surveys)

Works with small-to-medium datasets (a few hundred historical events)

Fast training & prediction (runs on a modest laptop)

Class imbalance (most events have <30% attendance)

XGBoost is not a black box—I can show my teacher exactly which features matter (e.g., 'distance from dorm to venue' or 'day before exam'). It's also what many Kaggle competitions use for tabular data, so it's career-relevant

Why XGBoost fits

XGBoost's native sparsity handling

Built-in regularization, cross-validation

Optimized C++ backend

scale_pos_weight parameter

DATA COLLECTION

Student features:

Distance from accommodation to event location (km)

Previous event attendance rate (last 3 events)

Academic load (number of exams in next 7 days – from student self-report or calendar)

Weather forecast (temp, snow, rain) – free API

Event features:

Day of week, start time

Event type (cultural, academic, sports, networking)

Lead time (how many days between RSVP and event)

Incentive (free meal? certificate? rare guest speaker?) – binary

Label (target):

attended = 1 if student checked in at event (via QR code or manual list), else 0.

Success Metrics

Technical: Achieve $AUC > 0.70$ on validation (reasonable for noisy human behavior data).

Practical: After 2 real events, show that predicted attendance is within 20% of actual (e.g., predicted 30, actual 25).

Community: Collect feedback from organizers: “Did this help you plan?” (Likert scale).

I will prove the model reduces over-ordering by at least 15% in a controlled test with two similar events – one using predictions, one not.

Why This is NOT Just "Another ML Project"

Most student ML projects predict house prices or Titanic survival. This project predicts human behavior in a culturally unique, resource-constrained environment – exactly the kind of problem ML should solve, but rarely does in coursework.

Cross-cultural relevance (African students in Russia → different attendance patterns than local students)

Deployment in a real platform (not a dead Jupyter notebook)

Ethical ML: predictions used to reduce waste, not manipulate people.

APPROVAL

I request approval to implement the Event Attendance Predictor using XGBoost as my coursework ML project, integrated into the live African Student Platform. I will deliver: code, a 5-minute demo, and a 2-page report on model performance and community feedback.

My grandmother always said, 'In our village, you cook for who actually comes, not who promises.' This ML feature finally lets our community do that – but in Vladivostok, in winter, with OVER 150 students."